You already have the hardest part:

✅ 10 years of data
✅ forecasting mindset
✅ feature engineering
✅ business understanding
✅ AI sentiment idea
✅ clear business use-case

Now the main challenge is execution order.

You should avoid trying to build everything at once.

YOUR BEST ROADMAP

You are building:

```
AI-powered egg market intelligence platform
```

So proceed in stages.

STAGE 1 — BUILD A RELIABLE FORECASTING CORE

Goal

Create a system that consistently predicts:

- next-day prices
- 7-day trends

before building fancy UI.

STEP 1 — FINALIZE DATA PIPELINE

Your current dataset is already close.

Now make a clean production pipeline.

Your Final Dataset Structure

Features

```
features = [
    "market_rating",

    "lag_1",
    "lag_3",
    "lag_7",
    "lag_14",
    "lag_30",

    "rolling_mean_7",
    "rolling_mean_14",
    "rolling_mean_30",

    "rolling_std_7",

    "tmax",
    "prcp",

    "dayofweek",
    "weekofyear",
    "quarter",
```

```
    "is_weekend",

    "is_festival"
]
```

Target

```
Price
```

STEP 2 — CREATE PROPER TRAINING PIPELINE

Make reusable functions.

Example:

```
load_data()
create_features()
train_model()
predict()
evaluate()
```

This is VERY important.

STEP 3 — TRAIN BASELINE MODEL

Start with:

- CatBoost

Why?

Because your data is:

- nonlinear
- seasonal
- sentiment-driven

CatBoost handles this beautifully.

STEP 4 — WALK-FORWARD VALIDATION

DO NOT use random split.

Use:

```
Train → future test
```

Example:

Train	Test
2015-2021	2022
2015-2022	2023
2015-2023	2024

This simulates real business forecasting.

STEP 5 — BUILD 7-DAY FORECAST ENGINE

Initially use:

```
recursive forecasting
```

Meaning:

```
predict day 1
↓
use prediction for day 2
↓
```

```
predict day 3
...
```

Much simpler initially.

STAGE 2 — CREATE BUSINESS KPIs

Before dashboard:

Define what businesses actually care about.

MOST IMPORTANT KPIs

Forecast KPIs

KPI	Meaning
tomorrow price	immediate planning
7-day avg	procurement
weekly trend	market direction
volatility	risk

Market KPIs

KPI	Meaning
bullish/bearish	sentiment
expected change %	movement

demand pressure	consumption
supply pressure	availability

Model KPIs

KPI	Meaning
MAE	avg error
RMSE	forecast quality
MAPE	percentage accuracy

STAGE 3 — BUILD POWER BI DASHBOARD

This should now become your business intelligence layer.

YOUR POWER BI DASHBOARD SHOULD HAVE

PAGE 1 — EXECUTIVE OVERVIEW

Cards:

- today price
- tomorrow prediction
- 7-day forecast
- sentiment
- volatility

PAGE 2 — PRICE ANALYTICS

Charts:

- historical trend
- predicted trend
- rolling averages
- forecast confidence

PAGE 3 — SEASONALITY

Show:

- monthly trends
- yearly behavior
- festival impact
- weekday patterns

This will become one of your strongest business insights.

PAGE 4 — CITY ANALYSIS

Compare:

City	Forecast	Trend	Risk

PAGE 5 — AI MARKET INTELLIGENCE

Show:

- market rating
- reasoning
- demand pressure
- expected movement

PAGE 6 — MODEL PERFORMANCE

Build trust.

Show:

- MAE
- RMSE
- prediction vs actual

STAGE 4 — AUTOMATION

This is where your project becomes real.

Daily automation:

```
Fetch latest prices
↓
Fetch weather
↓
Run AI agent
↓
Generate features
↓
Predict
↓
Save results
↓
Update dashboard
```

STAGE 5 — CREATE AI AGENT

This becomes your differentiator.

Your agent should analyze:

- poultry news
- feed prices
- weather
- festivals
- transportation
- local demand

and generate:

```
market_rating
expected_change_pct
market_summary
```

This is where your product becomes much stronger than simple forecasting systems.

STAGE 6 — BUILD WEB PLATFORM

ONLY after:

✅ model stable
✅ dashboard useful
✅ KPIs validated
✅ automation working

Then build:

- authentication
- API
- subscriptions
- alerts

- mobile-friendly UI

Use:

- FastAPI
- Next.js

MOST IMPORTANT TECHNICAL ADVICE

DO NOT OVERCOMPLICATE EARLY

Avoid:

❌ transformers
❌ advanced deep learning
❌ microservices
❌ Kubernetes
❌ distributed systems

Right now your priority is:

```
business value
```

not engineering complexity.

YOUR BIGGEST ADVANTAGE

Honestly:

your advantage is NOT the ML model.

It is:

`AI-assisted market reasoning`

Most people only forecast using historical prices.

You are combining:

- historical data
- weather
- seasonality
- sentiment
- AI intelligence

That is much more commercially valuable.

WHAT I WOULD PERSONALLY DO NEXT (IN ORDER)

WEEK 1

✅ finalize dataset
✅ finalize features
✅ build CatBoost model
✅ evaluate accuracy

WEEK 2

✅ build 7-day forecasting
✅ create KPI calculations
✅ automate prediction pipeline

WEEK 3

- ✅ build Power BI dashboard
- ✅ create business charts
- ✅ create seasonality analytics

WEEK 4

- ✅ build AI market sentiment agent
- ✅ generate market summaries
- ✅ add alerts

WEEK 5+

- ✅ backend APIs
- ✅ web platform
- ✅ user system
- ✅ mobile dashboard

FINAL THOUGHT

You are no longer building a small ML project.

You are building:

```
commodity intelligence infrastructure
```

If you execute carefully and keep the system simple + reliable initially, this can genuinely become a strong commercial analytics platform.